

during the climatic swings of the Ice Age, but population levels remained very low and the survival of early humans must have been precarious at times.

The Neanderthals

By 200,000 years ago, *Homo neanderthalensis* had evolved in Europe and Eurasia. The Neanderthals had large brains and more rounded heads than their predecessors. Their body shape was also more recognizably “human,” but it is believed that their reasoning power and speech were not as developed as those of *Homo sapiens*.

They were, however, expert hunters, who pursued animals such as bison with wooden and stone-tipped spears. They made sophisticated tools and dwelt in caves, rock shelters, and open camps. Theirs was a tough life in savage environments, and they probably lived for 30–40 years. Most experts agree that Neanderthals were not the ancestors of modern humans.

The appearance of modern humans

Intense controversy surrounds the origins of *Homo sapiens*—ourselves. Most geneticists use DNA evidence

(see pp.26–27) to argue that modern humans first appeared in tropical Africa by about 180,000 years ago. The earliest fully modern human fossils come from Huerto, Ethiopia, and date to about 160,000 years ago. From Africa, *Homo sapiens* spread across the Sahara and into southwestern Asia by 100,000 years ago. No one knows when humans developed the abilities that set them apart from their earlier ancestors, but they were fully developed by 45,000 years ago, when the first modern humans settled in Europe alongside the Neanderthals.

The arrival of *Homo sapiens* may have spelled the end for the Neanderthals.

EXTINCTION AND SUCCESS

Although Neanderthals and *Homo sapiens* lived alongside one another, DNA evidence suggests they did not interbreed. Neanderthals died out, perhaps at the hands of *Homo sapiens*, who were successful in adapting to every corner of the globe. More than any other species, humans have used their skills to their own advantage.

2.75–1 mya

Sites Africa

Brain size 500–550 cc

*Paranthropus boisei* is the most extreme version of the early “robust” humans living in eastern Africa. *Boisei* flourished in the dry savanna areas that existed in Africa at that time and may have died out after climate change.



Paranthropus boisei

2–0.5 mya

Site Africa, Asia, Europe

Brain size 810–1250 cc

*Homo erectus* was a powerfully built human with massive brow ridges, a large face, and a long, low skull to accommodate a much larger brain.



Homo erectus

350,000–24,000 ya

Site Africa and Eurasia

Brain size 1125–1550 cc

*Homo neanderthalensis* may have lived alongside modern *Homo sapiens* in Europe. The species had a large brain and short robust build with powerful limbs.



Homo neanderthalensis

1.5 MYA

1 MYA

0.5 MYA

PRESENT DAY

Homo ergaster



1.9–1.4 mya

Sites Africa

Brain size 600–910 cc

*Homo ergaster* was relatively tall, with a brain size well below that of modern humans. The skull was thick and the face long, with a “modern” projecting nose, a massive jawbone, and large teeth.



Homo heidelbergensis



600,000–250,000 ya

Sites Africa and Europe

Brain size 1225–1300 cc

*Homo heidelbergensis* may have been an ancestor of *Homo neanderthalensis* in Europe. The skull had a large brow ridge like *Homo ergaster* and *Homo erectus* but its brain was larger.



Homo sapiens



From 150,000 ya

Sites Worldwide

Brain size 900–2000 cc

*Homo sapiens* roughly translates as “wise man.” Our brain size is larger than earlier humans’, and it is perhaps this which has enabled us to thrive in a variety of environments around the world.

